Research Papers

Group 6

Title: GENAI: RAG USE CASES WITH VECTOR DB TO SOLVE THE LIMITATIONS OF LLMS

- **Author**: Sriramaraju Sagi
- **Year**: 2024
- **Journal**: International Journal of Computer Engineering & Technology
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ABSTRACT: This research delves into combining Retrieval Augmented Generation (RAG) with Vector Databases (Vector DB) to tackle the challenges faced by Language Models (LLMs) in Generative AI scenarios. Despite the progress made by LLMs in understanding and generating language issues such as data, hallucination and incorporating domain specific details persist. Our innovative method utilizes the semantically robust features of Vector DBs in conjunction with the RAG framework to improve aspects of LLM performance. By integrating time relevant information retrieved from Vector DBs LLMs can produce more precise, current and targeted content. We outline the procedures involved in gathering data from sources creating embeddings and assigning metadata to establish a repository that significantly enhances LLM generative capabilities. Our results suggest that this approach not only overcomes LLM limitations but also opens opportunities for their utilization, in fields requiring accuracy and timeliness.

Keywords: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Vector Databases (Vector DB), Semantic Embeddings

Tittle: Graph Retrieval-Augmented Generation: A Survey

- **Authors:** Boci Peng, Yun Zhu, Yongchao Liu, Xiaohe Bo, Haizhou Shi, Chuntao Hong, Yan Zhang, Siliang Tang
- **Year:** 2024
- **Journal: arXiv**
- **URL: https://arxiv.org/abs/2408.08921**

ABSTRACT: Retrieval-Augmented Generation (RAG) has achieved remarkable success in addressing the challenges of Large Language Models (LLMs) without necessitating retraining. By referencing an external knowledge base, RAG refines LLM outputs, effectively mitigating issues such as “hallucination”, lack of domain-specific knowledge, and outdated information. However, the complex structure of relationships among different entities in databases presents challenges for RAG systems. In response, GraphRAG leverages structural information across entities to enable more precise and comprehensive retrieval, capturing relational knowledge and facilitating more accurate, context-aware responses. Given the novelty and potential of GraphRAG, a systematic review of current technologies is imperative. This paper provides the first comprehensive overview of GraphRAG methodologies. We formalize the GraphRAG workflow, encompassing Graph-Based Indexing, Graph-Guided Retrieval, and Graph-Enhanced Generation. We then outline the core technologies and training methods at each stage. Additionally, we examine downstream tasks, application domains, evaluation methodologies, and industrial use cases of GraphRAG
Key Words: Large Language Models, Graph Retrieval-Augmented Generation, Knowledge Graphs, Graph Neural Networks

Title :HAKES: Scalable Vector Database for Embedding Search Service

- **Authors:** Guoyu Hu, Shaofeng Cai, Tien Tuan Anh Dinh, Zhongle Xie, Cong Yue, Gang Chen, Beng Chin Ooi
- **Year**: 2025
- **Journal:** arXiv
- **URL**: https://arxiv.org/abs/2505.12524

ABSTRACT: Modern deep learning models capture the semantics of complex data by transforming them into high-dimensional embedding vectors. Emerging applications, such as retrieval-augmented generation, use approximate nearest neighbor (ANN) search in the embedding vector space to find similar data. Existing vector databases provide indexes for efficient ANN searches, with graph-based indexes being the most popular due to their low latency and high recall in real-world high-dimensional datasets. However, these indexes are costly to build, suffer from significant contention under concurrent read-write workloads, and scale poorly to multiple servers. Our goal is to build a vector database that achieves high throughput and high recall under concurrent read-write workloads. To this end, we first propose an ANN index with an explicit two-stage design combining a fast filter stage with highly compressed vectors and a refine stage to ensure recall, and we devise a

novel lightweight machine learning technique to fine-tune the index parameters. We introduce an early termination check to dynamically adapt the search process for each query. Next, we add support for writes while maintaining search performance by decoupling the management of the learned parameters. Finally, we design HAKES, a distributed vector database that serves the new index in a disaggregated architecture. We evaluate our index and system against 12 state-of-the-art indexes and three distributed vector databases, using high-dimensional embedding datasets generated by deep learning models. The experimental results show that our index outperforms index baselines in the high recall region and under concurrent read-write workloads. Furthermore, HAKES is scalable and achieves up to 16× higher throughputs than the baselines.

URL: https://arxiv.org/abs/2505.12524

Title: RAGOps: Operating and Managing Retrieval-Augmented Generation Pipelines.

Authors: Xiwei Xu, Hans Weytjens , Dawen Zhang, Qinghua Lu, Liming Zhu, Ingo Weber

Year: **2025**

Journal: arXiv

url : https://arxiv.org/html/2506.03401v1#S7

Abstract

Recent studies show that 60% of LLM-based compound systems in enterprise environments leverage some form of retrieval-augmented generation (RAG), which enhances the relevance and accuracy of LLM (or other genAI) outputs by retrieving relevant information from external data sources. LLMOps involves the practices and techniques for managing the lifecycle and operations of LLM compound systems in production environments. It supports enhancing LLM systems through continuous operations and feedback evaluation. RAGOps extends LLMOps by incorporating a strong focus on data management to address the continuous changes in external data sources. This necessitates automated methods for evaluating and testing data operations, enhancing retrieval relevance and generation quality. In this paper, we (1) characterize the generic architecture of RAG applications based on the 4+1 model view for describing software architectures, (2) outline the lifecycle of RAG systems, which integrates the management lifecycles of both the LLM and the data, (3) define the key design considerations of RAGOps across different stages of the RAG lifecycle and quality trade-off analyses, (4) highlight the overarching research challenges around RAGOps, and (5) present two use cases of RAG applications and the corresponding RAGOps considerations.

Title : Automating Systematic Literature Reviews with Retrieval-Augmented Generation: A Comprehensive Overview

- **Year:** 2024
- **Journal:** Applied Sciences (MDPI)
- **Authors:** Binglan Han, Teo Susnjak, Anuradha Mathrani
- URL: **https://www.mdpi.com/2076-3417/14/19/9103**

Abstract

This study examines Retrieval-Augmented Generation (RAG) in large language models (LLMs) and their significant application for undertaking systematic literature reviews (SLRs). RAG-based LLMs can potentially automate tasks like data extraction, summarization, and trend identification. However, while LLMs are exceptionally proficient in generating human-like text and interpreting complex linguistic nuances, their dependence on static, pre-trained knowledge can result in inaccuracies and hallucinations. RAG mitigates these limitations by integrating LLMs' generative capabilities with the precision of real-time information retrieval. We review in detail the three key processes of the RAG framework—retrieval, augmentation, and generation. We then discuss applications of RAG-based LLMs to SLR automation and highlight future research topics, including integration of domain-specific LLMs, multimodal data processing and generation, and utilization of multiple retrieval sources. We propose a framework of RAG-based LLMs for automating SRLs, which covers four stages of SLR process: literature search, literature screening, data extraction, and information synthesis. Future research aims to optimize the interaction between LLM selection, training strategies, RAG techniques, and prompt engineering to implement the proposed framework, with particular emphasis on the retrieval of information from individual scientific papers and the integration of these data to produce outputs addressing various aspects such as current status, existing gaps, and emerging trends.

Title : Searching for Best Practices in Retrieval-Augmented Generation

- **Authors**: Xiaohua Wang, Zhenghua Wang, Xuan Gao, Feiran Zhang, Yixin Wu, Zhibo Xu, Tianyuan Shi, Zhengyuan Wang, Shizheng Li, Qi Qian, Ruicheng Yin, Changze Lv, Xiaoqing Zheng, Xuanjing Huang

- **Year**: 2024
- **Journal**: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*
- **URL**: https://aclanthology.org/2024.emnlp-main.981/

Abstract :Retrieval-augmented generation (RAG) techniques have proven to be effective in integrating up-to-date information, mitigating hallucinations, and enhancing response quality, particularly in specialized domains. While many RAG approaches have been proposed to enhance large language models through querydependent retrievals, these approaches still suffer from their complex implementation and prolonged response times. Typically, a RAG workflow involves multiple processing steps, each of which can be executed in various ways. Here, we investigate existing RAG approaches and their potential combinations to identify optimal RAG practices. Through extensive experiments, we suggest several strategies for deploying RAG that balance both performance and efficiency. Moreover, we demonstrate that multimodal retrieval techniques can significantly enhance question-answering capabilities about visual inputs and accelerate the generation of multimodal content using a “retrieval as generation” strategy.